

SRAI

EXERCISES AND COMPLETE SOLUTIONS

VECTOR FOUNDATIONS

From representation to reproducible high-dimensional reasoning

Production Unit	PU-B01-C03
Book / Lesson	Book 1 / Lesson 3
Subject	Vector Foundations
Control status	Approved controlled companion v1.0

Scientific Reasoning and Artificial Intelligence (SRAI)
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How to use this companion

Complete Part I before consulting Part II. Show definitions, intermediate calculations and interpretation. For computational exercises, use a clean runtime, preserve executable assertions and record every scaling, weighting and coordinate-order choice.

Learning outcomes

- Distinguish scalars, points and vectors and preserve coordinate meaning.
- Compute vector operations, norms, distances, dot products and cosine similarity.
- Derive and verify projections, rejections, orthogonality and orthonormal bases.
- Explain how scaling and weighting change the geometry of data.
- Implement safe vector operations with dimension and zero-vector checks.
- Use vectors responsibly in statistics, data science, embeddings and decision intelligence.

Assessment structure

Part	Focus	Points
A	Conceptual foundations	20
B	Mathematical reasoning	35
C	Reproducible computation	30
D	Applied decision system	15

PART I - Exercises

A. Conceptual foundations

Exercise 1. Scalar, point or vector? [4 points]

For each object, state whether it is best treated as a scalar, point, free vector or feature vector, and justify the coordinate system: annual rainfall; Dakar's mapped location; displacement from one survey cluster to another; a household profile (income, household size, distance to market).

Exercise 2. Coordinate order is part of the definition [4 points]

Two systems store the same district profile as $x=(\text{rainfall}, \text{fertilizer}, \text{market access})$ and $y=(\text{fertilizer}, \text{rainfall}, \text{market access})$. Explain why their arrays cannot be compared component by component until the schema is reconciled.

Exercise 3. Norm choice [4 points]

Explain what the L1, L2 and L-infinity norms emphasize. Give one professional situation in which each choice could be defensible.

Exercise 4. Similarity is not equivalence [4 points]

A document-retrieval model reports cosine similarity 0.94. State precisely what may be concluded and identify at least four reasons why 'the documents mean the same thing' is too strong.

Exercise 5. Representation controls [4 points]

List the minimum controls required before a multi-sector indicator vector can be compared across countries or years.

B. Mathematical reasoning**Exercise 6. Operations and norms [7 points]**

Let $x=(3,-4,12)$ and $y=(-1,2,0)$. Compute $x+y$, $x-y$, $2x$, $x \cdot y$, and the L1, L2 and L-infinity norms of x .

Exercise 7. Distance and scaling [7 points]

District A is represented by (1000 mm rainfall, 20 km market distance) and District B by (1100 mm, 5 km). Compute their raw Euclidean distance. Then express rainfall in metres and recompute. Explain why the substantive districts did not change even though the numerical distance did.

Exercise 8. Angle and cosine similarity [7 points]

For $x=(1,2,2)$ and $y=(2,1,2)$, compute the dot product, both Euclidean norms and cosine similarity. Determine whether the angle is acute, right or obtuse.

Exercise 9. Projection and rejection [7 points]

Let $x=(3,4)$ and $u=(1,1)$. Compute $\text{proj}_u(x)$, the rejection $r=x-\text{proj}_u(x)$, and verify independently that $u \cdot r=0$.

Exercise 10. Gram-Schmidt [7 points]

Apply Gram-Schmidt to $v_1=(1,1,0)$ and $v_2=(1,0,1)$. Produce q_1 and q_2 and verify $q_1 \cdot q_2=0$ and $\|q_1\|^2=\|q_2\|^2=1$.

C. Reproducible computation**Exercise 11. Safe normalization [7 points]**

Implement `normalize(x, tol=1e-12)`. It must accept a one-dimensional numeric array, reject non-finite values and raise a clear error for a zero or near-zero vector. Test ordinary, zero and invalid inputs.

Exercise 12. Cosine similarity from first principles [7 points]

Implement `cosine_similarity(x,y)` without calling a library cosine-similarity function. Validate equal dimensions and nonzero norms. Compare (1,0), (1,1), (-1,0) and (0,0) through assertions.

Exercise 13. Projection with invariants [8 points]

Implement `project(x,u)` and `reject(x,u)`. Validate dimensions and a nonzero direction. For at least three cases, assert $x=\text{project}(x,u)+\text{reject}(x,u)$ and $u \cdot \text{reject}(x,u) \approx 0$.

Exercise 14. Scaling sensitivity experiment [8 points]

Create a 5-by-3 synthetic district matrix with rainfall, fertilizer use and market distance. Compare pairwise distances under raw values, z-score standardization and one justified weighting scheme. Report at least one nearest-neighbour conclusion that changes.

D. Applied decision system

Exercise 15. National state-vector comparison [15 points]

Design a transparent comparison of at least six synthetic national or regional state vectors.

- Declare the unit of analysis, coordinate order, units, indicator direction and reference period.
- Define missing-data treatment and preserve uncertainty or quality flags.
- Compare raw, standardized and weighted representations.
- Use both Euclidean distance and cosine similarity and explain what each omits.
- Return results with reasons, transformation version and reproducibility metadata.
- State three safeguards against over-interpreting geometric similarity as equal welfare, causation or policy interchangeability.

PART II - Complete solutions

Solution 1. Scalar, point or vector?

Annual rainfall is a scalar when it is one magnitude for one unit and period. Dakar's mapped location is a point in a declared coordinate reference system. Displacement is a vector because it represents change in location with magnitude and direction. A household profile is a feature vector: ordered components whose meaning depends on a data schema. A point can be subtracted from another point to form a displacement vector; treating coordinates without their reference system is incomplete.

Solution 2. Coordinate order is part of the definition

The arrays occupy different bases or schemas: the first component of x is rainfall, while the first component of y is fertilizer. Componentwise subtraction would mix unlike variables. Reconcile the field names, units and order, then reorder one array before computing any norm, distance or similarity. The schema is part of the mathematical object, not merely metadata.

Solution 3. Norm choice

The L1 norm sums absolute component magnitudes and is relatively robust to one unusually large coordinate; it is useful when additive absolute deviation has meaning. The L2 norm gives Euclidean length and is natural for geometric energy and least-squares settings. The L-infinity norm records only the largest absolute coordinate and is useful when the worst component controls risk or tolerance. Each interpretation still requires comparable units or an explicit transformation.

Solution 4. Similarity is not equivalence

A value of 0.94 says that two nonzero embedding vectors point in similar directions under a particular representation. It does not establish identical meaning, factual agreement or causal connection. The value depends on the embedding model, training data, tokenization, pooling, language and domain, preprocessing and possible loss of negation or context. Task-specific validation and human inspection remain necessary.

Solution 5. Representation controls

Required controls include: coordinate names and order; units and reference periods; population and data provenance; missing-value rules; indicator direction; scaling and normalization; weights and their justification; uncertainty or quality flags; transformation version; sensitivity analysis; and review of whether the chosen similarity measure answers the decision question.

Solution 6. Operations and norms

$x+y=(2,-2,12)$; $x-y=(4,-6,12)$; $2x=(6,-8,24)$. The dot product is $3(-1)+(-4)(2)+12(0)=-11$. For x , $\|x\|_1=3+4+12=19$; $\|x\|_2=\sqrt{9+16+144}=13$; and $\|x\|_\infty=12$.

Solution 7. Distance and scaling

In the original units, $B-A=(100,-15)$, so the distance is $\sqrt{10000+225}=\sqrt{10225}\approx 101.119$. Expressing rainfall in metres gives $B-A=(0.1,-15)$, so the distance becomes $\sqrt{0.01+225}\approx 15.0003$. Only the numerical representation changed. This demonstrates that unstandardized Euclidean distance is not invariant to changes of measurement unit and can be dominated by a large numerical scale.

Solution 8. Angle and cosine similarity

$x \cdot y = 2 + 2 + 4 = 8$. Both vectors have norm $\sqrt{1+4+4} = 3$. Therefore cosine similarity is $8/9 \approx 0.8889$. Because the dot product is positive, the angle is acute (approximately 27.27 degrees).

Solution 9. Projection and rejection

$u \cdot u = 2$ and $x \cdot u = 7$, so $\text{proj}_u(x) = (7/2)(1,1) = (3.5, 3.5)$. The rejection is $r = (-0.5, 0.5)$. The independent check gives $u \cdot r = -0.5 + 0.5 = 0$, establishing orthogonality.

Solution 10. Gram-Schmidt

$q_1 = v_1 / \|v_1\| = (1/\sqrt{2}, 1/\sqrt{2}, 0)$. Remove the q_1 component from v_2 : $w_2 = v_2 - (v_2 \cdot q_1)q_1 = (1, 0, 1) - (1/2, 1/2, 0) = (1/2, -1/2, 1)$. Its norm is $\sqrt{3/2}$. Thus $q_2 = (1/\sqrt{6}, -1/\sqrt{6}, 2/\sqrt{6})$. Direct calculation gives $q_1 \cdot q_2 = 0$, and both squared norms equal 1.

Solution 11. Safe normalization

A safe implementation validates dimensionality, finiteness and the norm before division. The tolerance prevents numerically negligible vectors from producing unstable directions.

```
import numpy as np

def normalize(x, tol=1e-12):
    x = np.asarray(x, dtype=float)
    if x.ndim != 1 or not np.all(np.isfinite(x)):
        raise ValueError("x must be a finite one-dimensional vector")
    n = np.linalg.norm(x)
    if n <= tol:
        raise ValueError("cannot normalize a zero or near-zero vector")
    return x / n

assert np.allclose(normalize([3, 4]), [0.6, 0.8])
for bad in ([0, 0], [1, np.nan]):
    try: normalize(bad)
    except ValueError: pass
    else: raise AssertionError("expected ValueError")
```

Solution 12. Cosine similarity from first principles

The dot product is divided by the product of the two norms. Equal dimension and nonzero-domain checks are substantive preconditions, not optional defensive programming.

```
def cosine_similarity(x, y):
    x, y = np.asarray(x, float), np.asarray(y, float)
    if x.ndim != 1 or y.ndim != 1 or x.shape != y.shape:
        raise ValueError("vectors must be one-dimensional and equal in length")
    nx, ny = np.linalg.norm(x), np.linalg.norm(y)
    if nx <= 1e-12 or ny <= 1e-12:
        raise ValueError("cosine similarity is undefined for a zero vector")
    return float(np.dot(x, y) / (nx * ny))

assert np.isclose(cosine_similarity([1,0],[1,1]), 1/np.sqrt(2))
assert np.isclose(cosine_similarity([1,0],[-1,0]), -1)
```

Solution 13. Projection with invariants

The decomposition and orthogonality assertions verify defining properties independently of a displayed numeric result.

```
def project(x, u):
    x, u = np.asarray(x,float), np.asarray(u,float)
    if x.ndim != 1 or u.ndim != 1 or x.shape != u.shape:
        raise ValueError("vectors must have equal one-dimensional shape")
    den = float(np.dot(u,u))
    if den <= 1e-24:
        raise ValueError("projection direction must be nonzero")
    return (float(np.dot(x,u))/den) * u

def reject(x, u):
    return np.asarray(x,float) - project(x,u)

for x,u in [( [3,4],[1,1] ), ( [2,-1],[0,2] ), ( [1,2,3],[1,0,1] )]:
    p, r = project(x,u), reject(x,u)
    assert np.allclose(np.asarray(x,float), p+r)
    assert np.isclose(np.dot(u,r), 0.0, atol=1e-10)
```

Solution 14. Scaling sensitivity experiment

One valid solution constructs a labelled DataFrame, calculates distances with `scipy.spatial.distance` or NumPy, then repeats after `StandardScaler` and after applying documented square-root weights. The report should show the matrices or nearest-neighbour ranks. A changed nearest neighbour is not an error: it is evidence that geometry depends on the transformation. The defensible conclusion identifies which representation corresponds to the intended decision and reports sensitivity rather than selecting the most convenient result.

Solution 15. National state-vector comparison

A complete solution begins with a data dictionary and versioned transformation specification. It sends incomplete records to REVIEW or applies a justified imputation with flags; reverses undesirable indicators before interpretation; fits scaling parameters on the declared reference population; and records weights. Euclidean distance describes transformed separation including magnitude, whereas cosine similarity emphasizes direction and omits magnitude. Results must include the representation version, data period, quality flags and sensitivity comparison. Safeguards include expert validation of indicators and weights, uncertainty reporting, subgroup and perturbation checks, and a prohibition on interpreting geometric proximity as causal evidence or automatic policy equivalence.

Reproducibility checklist

- Run the canonical notebook from a clean kernel or Colab runtime.
- Record Python and package versions and preserve executable assertions.
- Declare vector dimension, coordinate order, units, scaling, weighting and missing-data rules.
- Test zero vectors, mismatched dimensions, non-finite values and near-degenerate inputs.
- Verify projection with $u \cdot r = 0$ and orthonormal bases with $Q \cdot \text{transpose } Q = I$.
- Complete sensitivity analysis before updating checksums and the controlled release record.

Alignment record

This companion is derived from PU-B01-C03_Chapter3_Vector_Foundations_Audited_Controlled_Edition_v1.0 and M1_N03_vector_foundations.ipynb. It expands the chapter assessment into a standalone 100-point resource while preserving SRAI Principle No. 3: geometry becomes evidence only after representation, scaling and interpretation have been justified.

Controlled references

- NumPy documentation: Linear algebra and `numpy.linalg.norm`.
- SciPy documentation: Spatial distance functions.
- scikit-learn documentation: `preprocessing.StandardScaler` and pairwise metrics.
- SRAI Book 1, Chapter 3: Vector Foundations (controlled edition v1.0).